Segment-4

Structured Analysis: Structured analysis is a development method that allows the analyst to understand the system and its activities in a logical way.

It has following attributes –

① It is graphical which specifies the presentation of application.

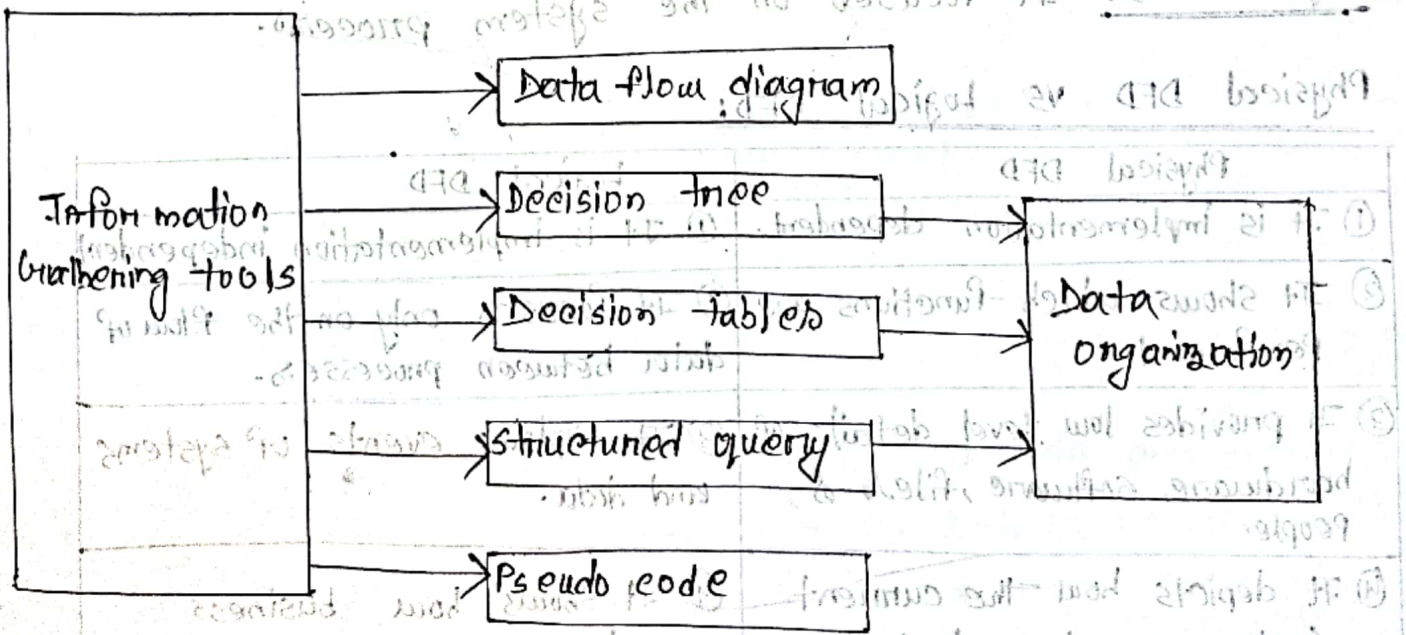
② It divides the processess so that it gives a clear picture of system flow.

③ It is logical rather than physical.

④ It is an approach that works from high-level overviews to lower-level details.

Structured analysis tools:

① Data-flow diagrams
② Data dictionary
③ Decision trees
④ Decision tables
⑤ Structure english.
⑥ Pseudocode



Data flow diagram (DFD) or bubble chart:

Data flow diagram is a graphical representation of flow of data in an information system.

⇒ It shows the flow of data between various function of system.
⇒ It specifies how the current system is implemented.
⇒ It is an initial stage of design phase.
⇒ Its graphical nature makes it a good communication tool between user and analyst.
⇒ It gives an overview of what data a system processes, what data are stored, what results are produced.

There are two types of DFD. They are, ① Logical DFD
② Physical DFD

- **Physical DFD:** It shows how the data flow is actually implemented in system.

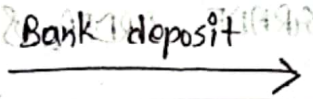
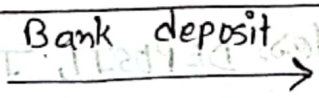
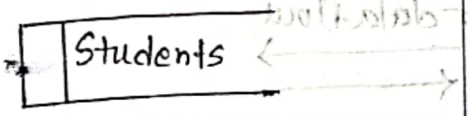
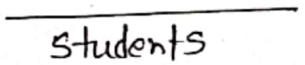
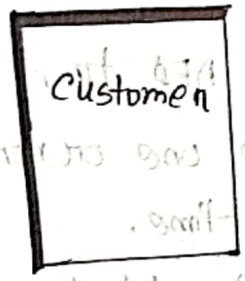
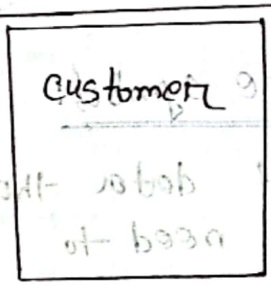
Logical DFD: It focused on the system process.

Physical DFD vs Logical DFD:

Physical DFD	Logical DFD
① It is implementation dependent.	① It is implementation independent
② It shows which functions are performed.	② It focuses only on the flow of data between processes.
③ It provides low level details of hardware, software, files & people.	③ It explains events of systems and data.
④ It depicts how the current system operates and how a system will be implemented.	④ It shows how business operates, not how the system can be implemented.

DFD Symbols: DFD use basic four symbols that represent processes, data flows, data stores and entities.

There are two types of symbol set. They are, ① Gane & Sarson ② Yourdon symbol set

Gane and Sarson	Symbol name	Yourdon symbol
	Process	
	Data flow	
	Data store	
	External entity	

Four basic DFD symbol: ① Processes Symbol
② Data Flow symbol
③ Data store symbol
④ Entities Symbol

① Processes Symbol: A process receives input data and produces output has a different content, form or both. The process calculating for pay uses two inputs to produce one output.

⇒ Processes contain business logic, also called business rules that

transform the data and produce the required results.

Examples: APPLY RENT PAYMENT, CALCULATE COMMISSION, ASSIGN FINAL GRADE, VERIFY ORDER and FILL ORDER.

Process

② Data flow symbol: A data flow is a path for data to move from one part of the information system to another.

⇒ It consists of singular noun and adjective.

Examples: DEPOSIT, INVOICE PAYMENT, STUDENT GRADE, ORDER COMISSION.

data flow

③ Data store symbol: A data store is used in a DFD to represent data that the system stores because one or more processes need to use the data as a later time.

⇒ A DFD does not show the detailed contents of a data store.

⇒ One incoming & one outgoing data flow is connected to a process.

Examples: STUDENTS, ACCOUNTS RECEIVABLE, PRODUCTS, DAILY PAYMENTS

Data Store

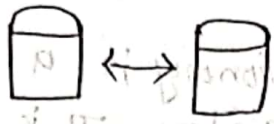
④ Entity symbol: DFD shows only external entities that provide data to the system or receive output from the system.

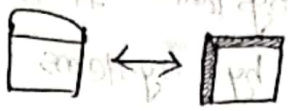
⇒ DFD entities also are called terminations, because they are data origins or final destinations.

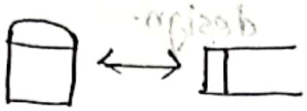
Entity

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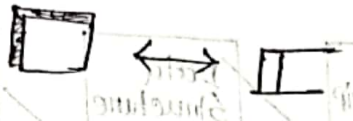
Rules for DFD symbol:

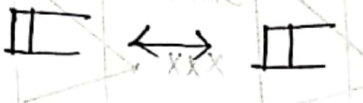

(Process to process) ✓


(Process to external entity) ✓


(Process to data store) ✓


(External entity to external entity) ✗


(external entity to data store) ✗


(Data store to data store) ✗

Guidelines for drawing DFD:

1. Draw the context diagram so it fits on one page.
2. Use the name of the information system as the process name in the context diagram.
3. Use unique names within each set of symbols.
4. Do not cross lines.
5. Provide a unique name and reference number for each processes
6. Obtain as much user input and feedback as possible.

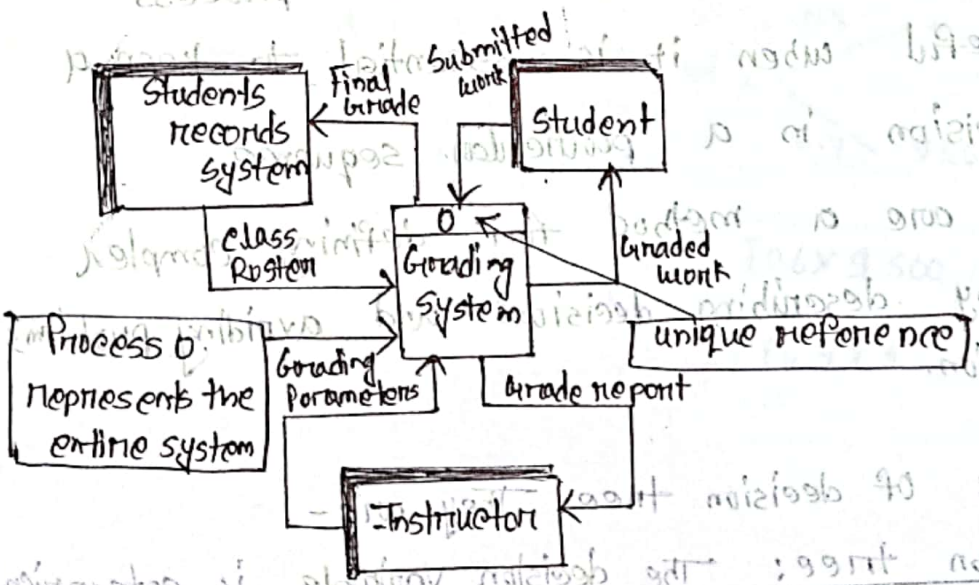


Fig: Context diagram DFD for a grading system

Data dictionary: A data dictionary is a structured repository of data elements in the system. It is a reference work of data and one is compiled by systems analysts to guide them through analysis and design.

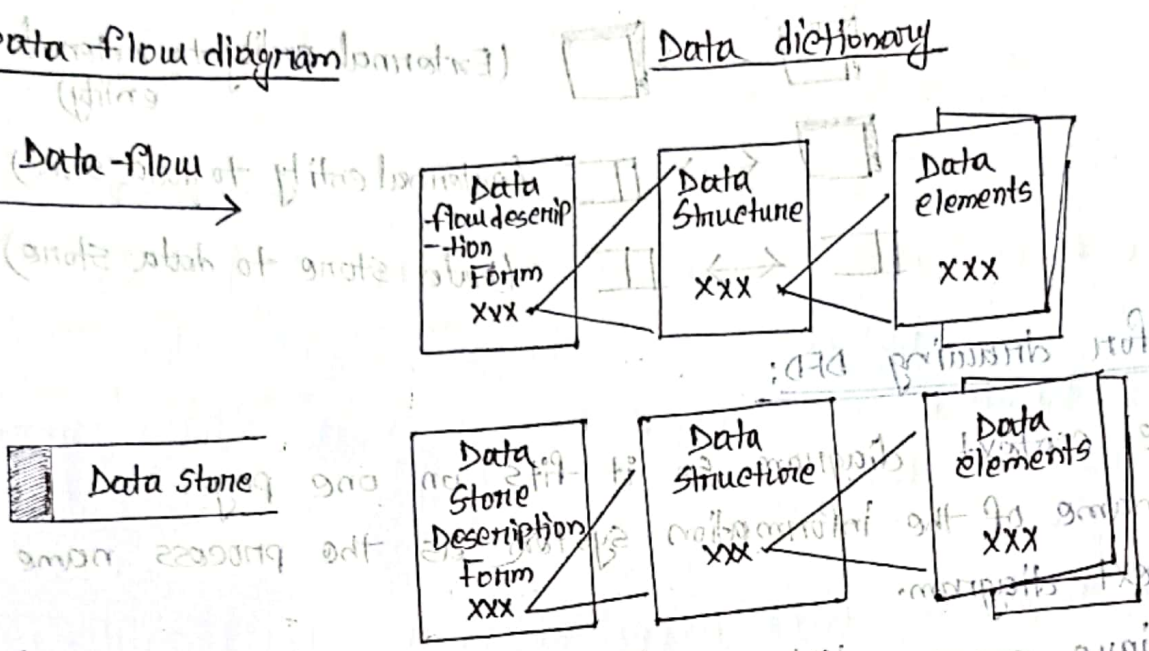


Fig: How data dictionaries relate to data flow diagram.

Decision Trees: Decision trees are used when complex branching occurs in a structured decision process.

⇒ Trees are useful when it is essential to keep a string of decision in a particular sequence.

⇒ Decision trees are a method for defining complex relationships by describing decisions and avoiding problems in communication.

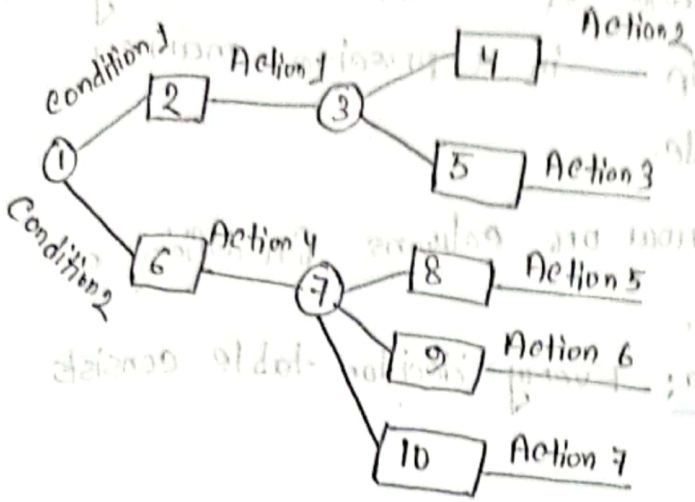
There are two types of decision tree. They are -

① **Classification decision tree:** The decision variable is categorical.

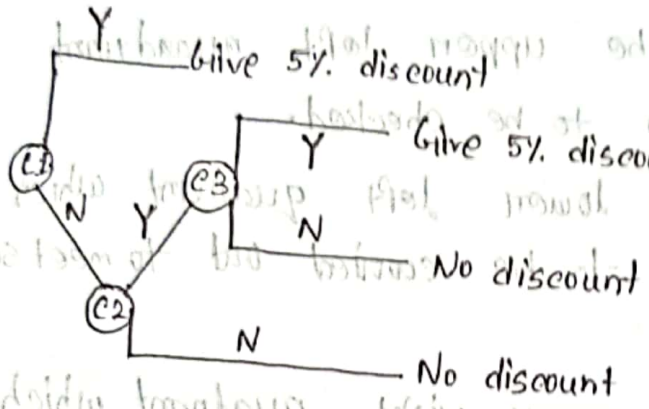
Ex: The above decision tree.

② Regression decision tree: The decision variable is continuous.

①



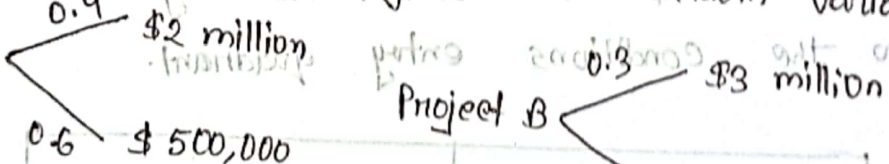
②



Y = Yes, N = No
C1: Advance payment mode
C2: Purchase Amount >= 10,000
C3: Regular customer

- $EV = \sum P(X_i) \times X_i$

where: x is a random variable
P(x) is the probability of the random variable



EV (Project A) = [0.4 × $2,000,000] + [0.6 × $ 500,000] = $ 1,100,000

EV (Project B) = [0.3 × $3,00,000] + [0.7 × $ 200,000] = $ 1,040,000

Decision table: Decision tables are a method of describing the complex logical relationship in a precise manner which is easily understandable.

⇒ It is a matrix containing row or columns for defining a problem and the actions.

Components of decision table: Every decision table consists of 4 parts:

① **Condition stub** - It is in the upper left quadrant which lists all the condition to be checked.

② **Action stub** - It is in the lower left quadrant which outlines all the actions to be carried out to meet such condition.

③ **Condition Entry** - It is in upper right quadrant which provides answers to questions asked in condition stub quadrant.

④ **Action Entry** - It is in lower right quadrant which indicates the appropriate action resulting from the answers to the conditions entry quadrant.

Conditions & actions	Rules
Conditions	Condition Alternatives
Actions	Action entries

Example - How to create a login screen Decision Base table

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Username (T/F)	F	T	F	T
Password (T/F)	F	F	T	T
Output (E/H)	E	E	E	H

Legend, T - Login and password are connect
F - Incorrect login or password
E - Error message appears.
H - Home screen appears

Structured english: Structured english is derived from structured Programming language which gives more understandable & precise description of process.

~~→ Structured english is derived from structured programming language.~~

Structure english type:

① **Sequential structure:** Here we set action & no branching occurs.

Example - Action #1
Action #2
Action #3

② **Decision structure:** If a condition is true complete the following statements otherwise jump to the else.

Ex: IF Condition A is true then implement Action A
ELSE implement Action B
ENDIF

③ **Case structure:** A special type of decision in which the cases are unique.

Ex: IF case #1 implement Action #1
ELSE IF case #2
Implement Action #2
ELSE IF case #3
Implement Action #3
ELSE IF case #4
Implement Action #4
ELSE Print error
ENDIF

④ Iteration: Blocks of statements that are repeated until done.

Ex: DO WHILE there are customers.
Action #1.
ENDDO

Pseudocode: A pseudocode does not confirm to any programming language and expresses logic in plain english.

- ⇒ It may specify the physical programming logic without actual coding during & after the physical design.
- ⇒ It is used in conjunction with structured programming.
- ⇒ It replaces the flowcharts of a program.

Guidelines for selecting appropriate tools:

1. Use DFD at high or low level analysis for providing good system.
2. Use data dictionary to simplify the structure.
3. Use structured english if there are many loops & actions are complex.
4. Use decision tables.
5. Use decision trees.

Developing Data flow diagram: Step 1: Draw a Context diagram
Step 2: Draw a diagram 0 DFD
Step 3: Draw the Lower-Level Diagrams

Step 1: A context diagram is a top level view of an information system that shows the systems boundaries and ~~system~~ scope. To draw a context diagram we have to use a single process symbol & identify it as process 0.

Step 2: To show the detail inside black box, we have to create DFD diagram 0. Diagram 0 zooms in on the system and shows major internal processes, data flows & data stores.

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Diagram 0 also repeats the entities and data flows that appear in the context diagram. When you expand the context diagram into DFD diagram 0, you must retrain all the connections that flow into and out of process 0.

Step 3: The set of lower-level DFD is based on the order system. To create lower-level diagrams, you must use leveling and balancing techniques. There are two techniques.

① **Leveling technique** – Leveling is the process of drawing a series of increasingly detailed diagram until all functional primitives are

② **Balancing technique** – Balancing maintains consistency among a set of DFDs by ensuring that input and output data flows align properly.

Example-1 (order system)

Context diagram for an order system:

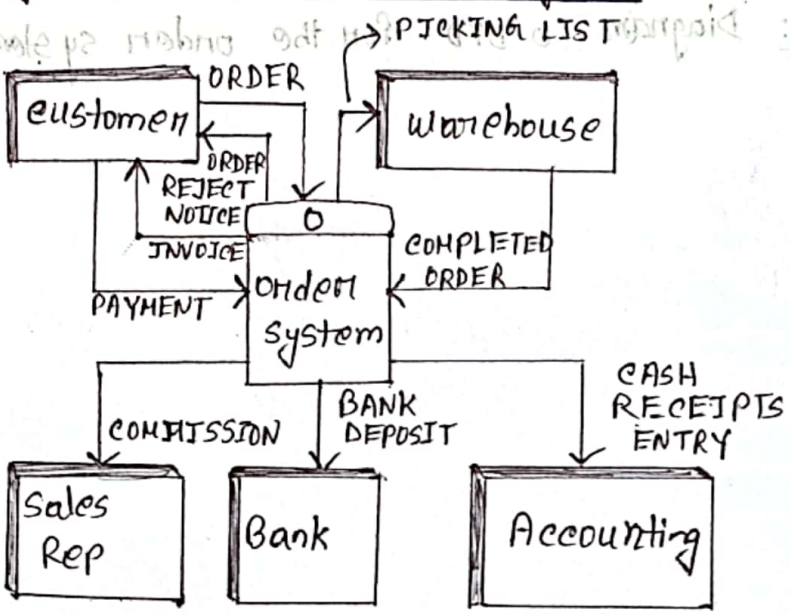


Fig: Context diagram DFD for an order system

Diagram 0 DFD:

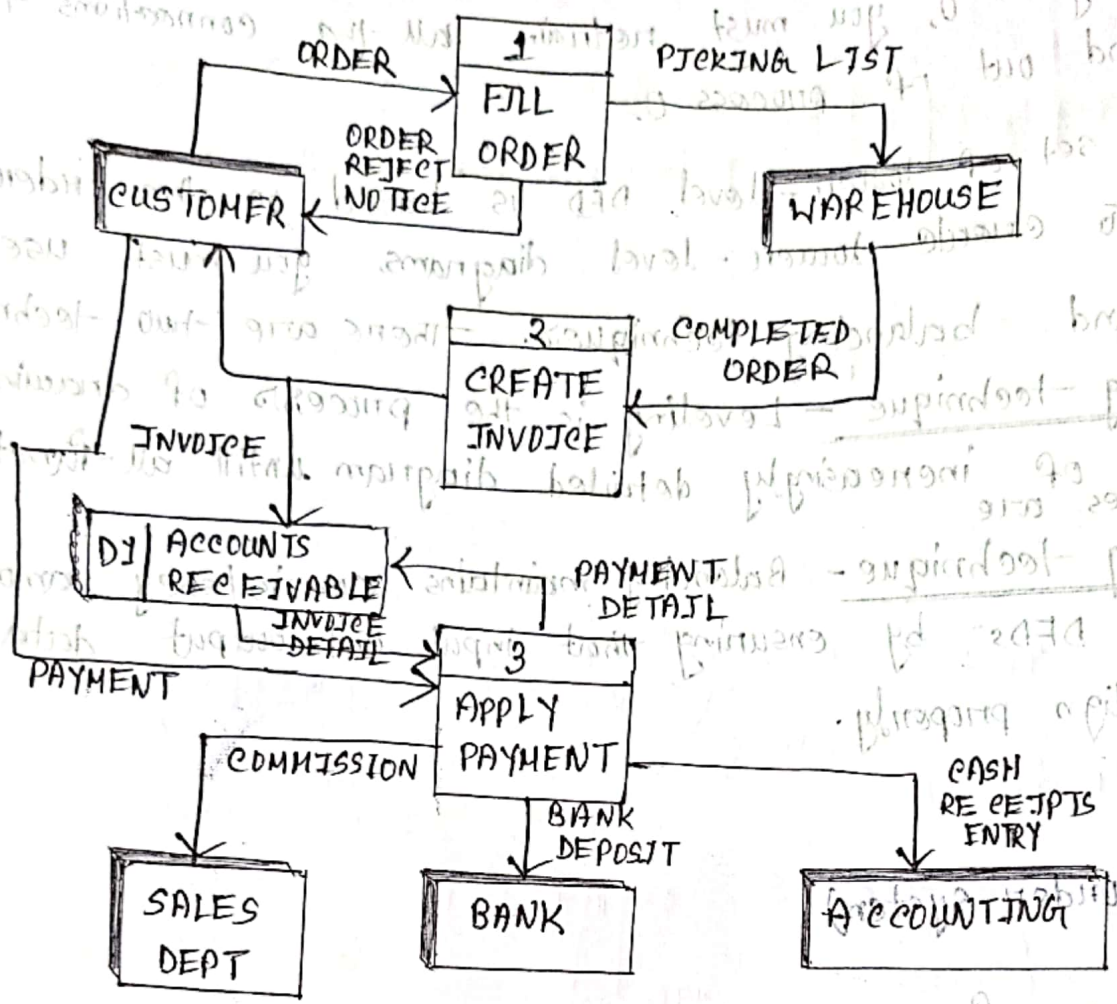
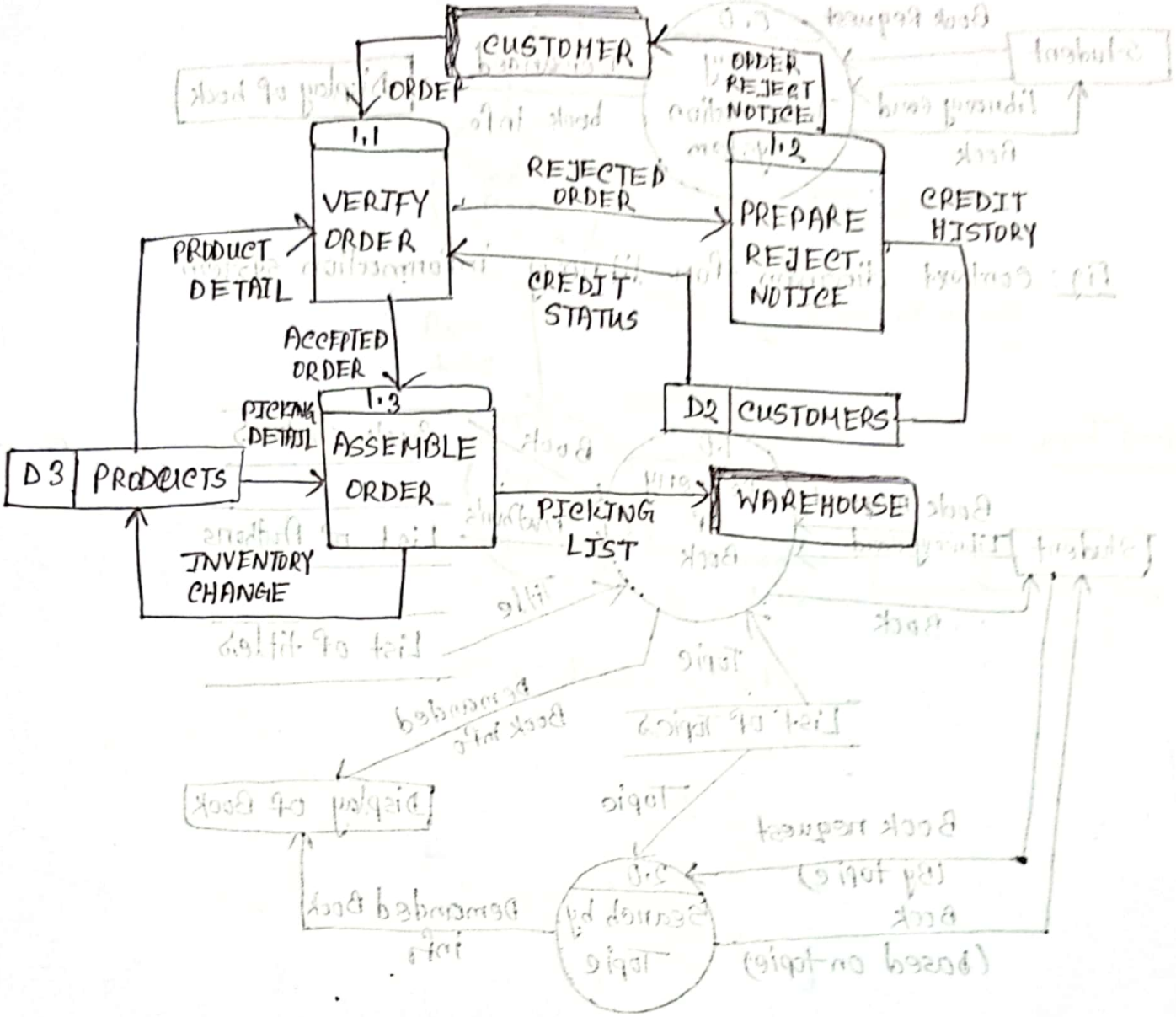


Fig: Diagram 0 DFD for the order system

Draw the lowest level diagrams



Example-2 (Library Management System)

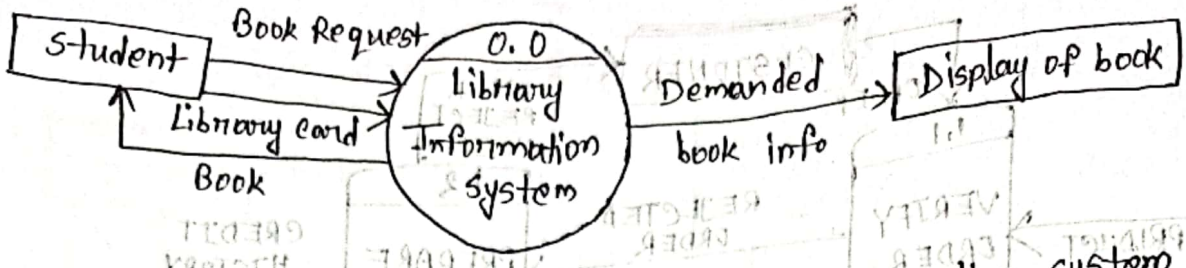


Fig: context diagram for library information system

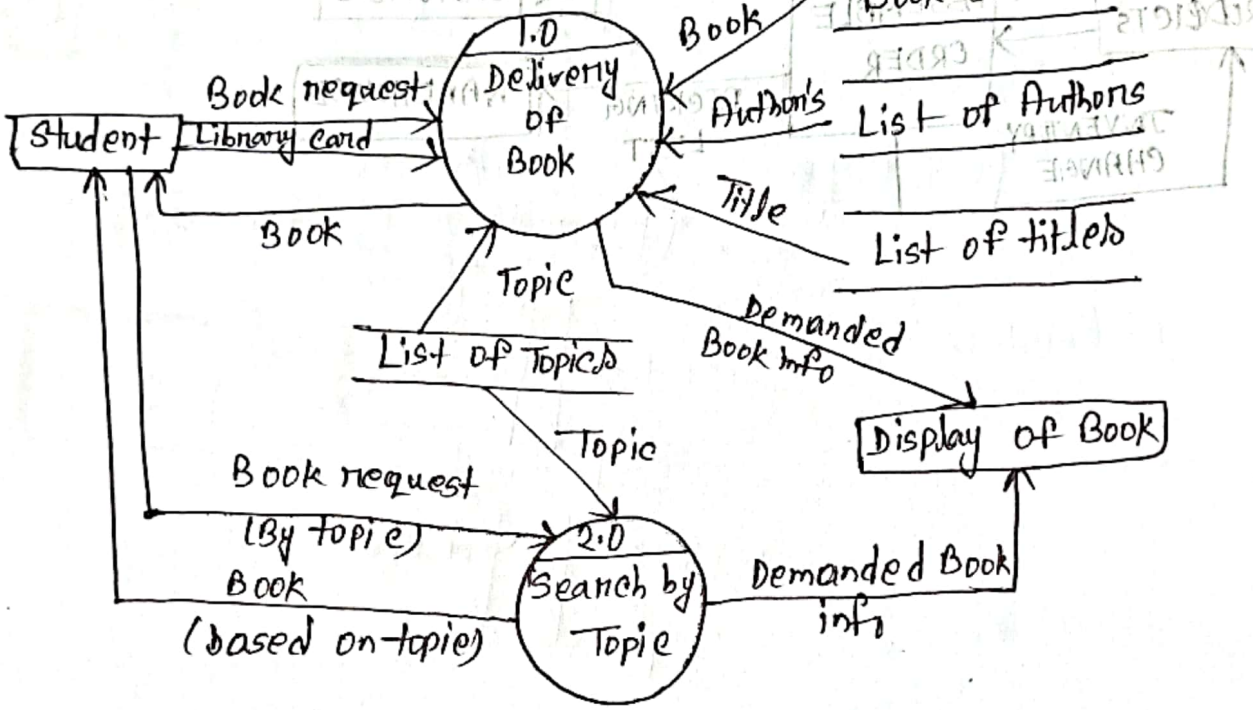


Fig: Level 1 DFD

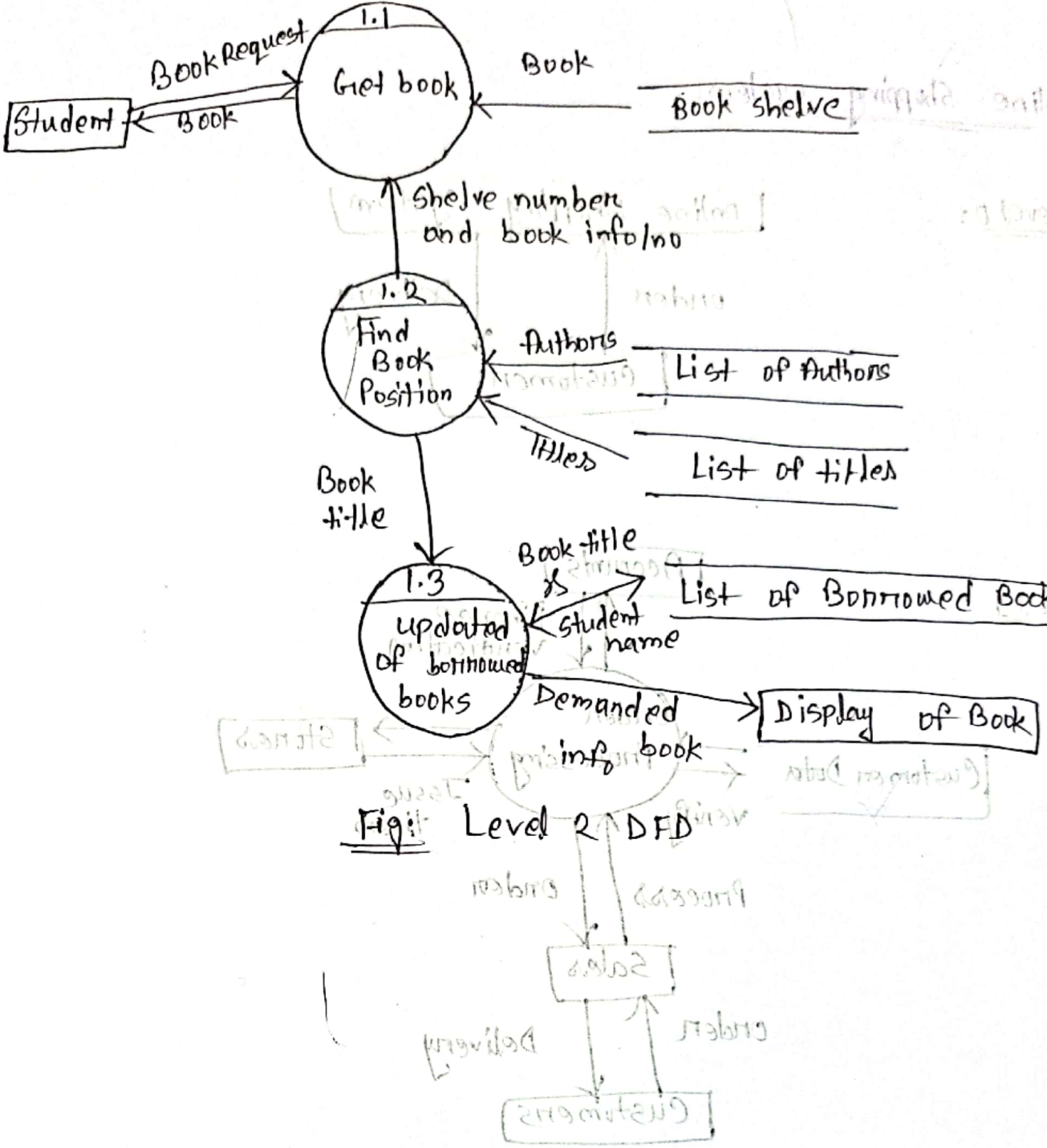


Fig: Level 2 DFD